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Preliminary results of a palaeomagnetic and rock magnetic study of the Proterozoic Tsuomasvarri intrusions, northern Fennoscandia

S. Mertanen, L.J. Pesonen

Laboratory for Palaeomagnetism, Geological Survey of Finland, SF-02150 Espoo, Finland

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Abstract

Preliminary palaeomagnetic results are presented for the Proterozoic Tsuomasvarri central ultramafic intrusion and surrounding, older gabbro-diorite intrusion in the northeastern Fennoscandian shield. In the ultramafic intrusion, five remanence components, reflecting different geological processes, were isolated in the alternating field (AF) and thermal demagnetization of 55 samples from ten sites. A reversed polarity remanence component C (pole: Plat=26°N, Plon=245°E, dp=5°, dm=9°, n=32 samples) dominates in the ultramafic intrusion. The normal polarity pole A (Plat=44°N, Plon=232°E, dp=4°, dm=6°, n=9 samples) corresponds to an age of ~1.88 Ga on the Apparent Polar Wander Path (APWP), and is probably a thermochemical remanence acquired during the Svecokarelian orogeny or in the late stages of magma cooling. Pole B (Plat=35°N, Plon=158°E, dp=9°, dm=15°, n=5 samples, normal polarity) corresponds to the ~1.75 Ga old post-orogenic activity in the area, and is also seen in results of K–Ar (whole rock) isotopic age determinations from the vicinity. The normal polarity pole D (pole: Plat=25°S, Plon=285°E, dp=12°, dm=19°, n=11 samples) has a palaeomagnetic age of ~2.44 Ga which is in contrast with the isotopic age determination of 1.93 Ga (U–Pb age on zircon) of the gabbro-diorite intrusion. Pole F (Plat=3°N, Plon=212°E, dp=10°, dm=18°, n=5 samples) indicates an age of 0.5–0.4 Ga on the Palaeozoic APWP segment of Fennoscandia and probably reflects a weak Caledonian overprint in northeastern Finland.

In the gabbro-diorite intrusion, two remanence components were isolated in 46 samples from eight sites. The palaeomagnetic age, ~1.93 Ga, of the dominant normal polarity component, A′ (pole: Plat=40°N, Plon=247°E, dp=5°, dm=8°, n=28 samples) is consistent with the isotopic age of 1.93 Ga (U–Pb determination on zircon). The normal polarity component D (pole: Plat=20°S, Plon=285°E, dp=14°, dm=23°, n=5 samples), which was also isolated in the ultramafic intrusion, contradicts the U–Pb isotopic age determination.

1. Introduction

In the last decade the northeastern Fennoscandian shield has been a subject of keen interest among those seeking to understand the formation of Archaean and Proterozoic crust. Models have been proposed for the geotectonic evolution of the crust (e.g. Hörmann et al., 1980; Barbey et al., 1984; Krill, 1985; Marker, 1985, 1990; Berthelsen and Marker, 1986; Barbey and Martin, 1987; Gaál and Gorbatschev, 1987). In the European Geotraverse Project (EGT), the structure and evolution of the Fennoscandian lithosphere were studied through integrated geologic and geophysical studies (Freeman et al., 1989). It was suggested by Marker (1985, 1990) and Berthelsen and Marker (1986) that the Kola suture, running from northern Norway and Fin-

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land to Kola Peninsula in Russia (Figs. 1 and 2), represents a 2.4–1.9 Ga old collision belt between two Archaean cratons: the Inari Terrain in the southwest and the Sorvaranger Terrain in the northeast. However, recent geological and isotopic age determinations (Hanski and Smolkin, 1989; Hanski et al., 1990; Kesola, 1991) from this area show that the Kola suture belt consists

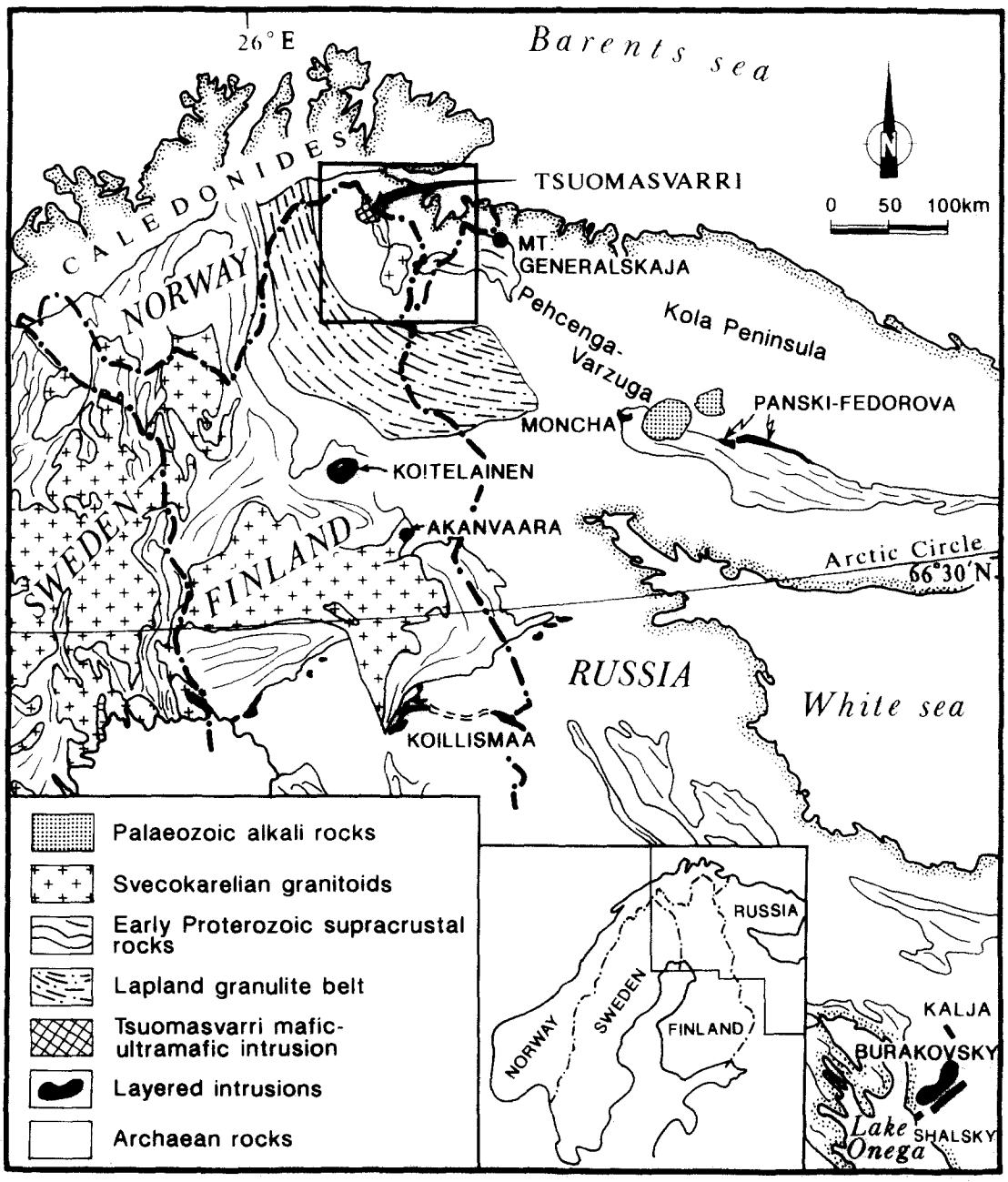


Fig. 1. Generalized geological map of the northeastern Fennoscandian shield. The map is mainly based on Alapieti and Lahtinen (1989) and other sources. The inset is shown in Fig. 2.

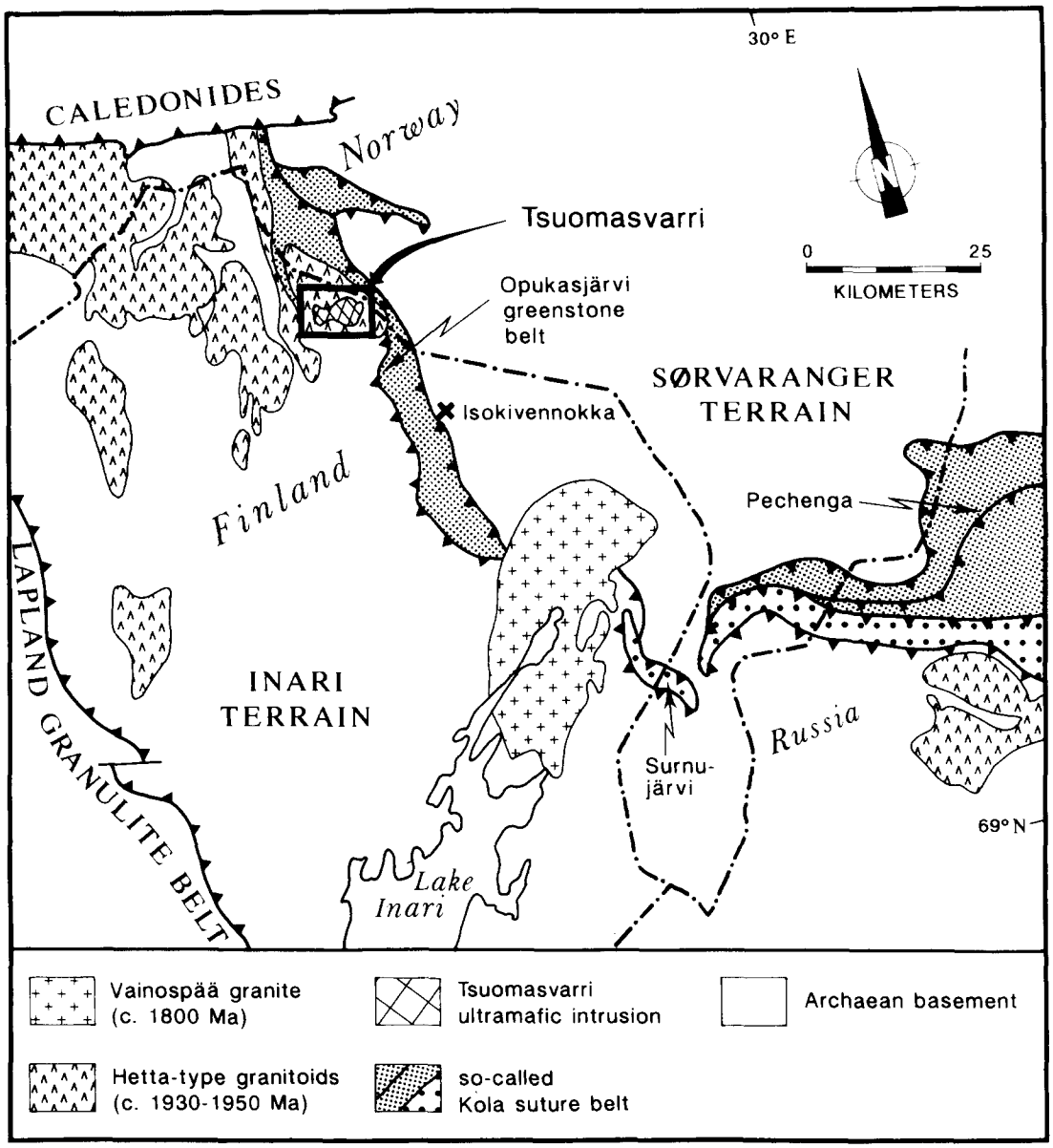


Fig. 2. Generalized geological map of the study area (inset in Fig. 1) (after Berthelsen and Marker, 1986, and Kesola, 1992).

of different rift-related parts of varying ages and thus, there is no uniform belt formed in a single collisional or plate-tectonic event.

Palaeomagnetism allows plate movements between blocks to be measured by comparing age-equivalent natural remanent magnetization (NRM) directions in both sides of a collision belt. If the NRM directions in rocks older than the proposed collision event diverge, relative movement has taken place between the blocks. Initially, one of the aims of our palaeomagnetic study was to test the proposed plate tectonic models associated with the Kola suture belt. For this reason the Tsuomasvarri ultramafic intrusion and the surrounding gabbro-diorite intrusion located at the southwestern margin of this

belt (Figs. 1 and 2) were selected for a palaeomagnetic study. On geological grounds it was suggested that the Tsuomasvarri ultramafic intrusion was Palaeoproterozoic in age (~2.4 Ga) (R. Kesola, pers. commun., 1986). The Archaean Isokivennokka granite (~2.6 Ga) (Fig. 2) was chosen to represent the northeastern side of the Kola suture. Later, however, new isotopic age determinations (M. Vaasjoki, in Forsberg-Heikkilä, 1989) showed that the gabbro-diorite is as young as of Svecokarelian age (1.93 Ga) and since it cuts the ultramafic intrusion, the ultramafic intrusion is younger than 1.93 Ga. Furthermore, samples from the Isokivennokka granite did not give reliable palaeomagnetic results. Hence, plate tectonic interpretations of the Kola suture belt could not be tested with palaeomagnetic data.

Palaeomagnetic investigations were carried out on the Tsuomasvarri layered ultramafic intrusion and the surrounding gabbro-diorite intrusion (Fig. 3), because they proved to carry multicomponent remanent magnetizations (Mertanen and Pesonen, 1992). It is important to isolate the different magnetizations, because they can have been formed in several thermal, chemical or hydrothermal processes and may thus reflect the complex geological evolution of the northern Fennoscandian shield. The aim of this paper is to study the nature and origin of these magnetizations. We have attempted to associate the magnetizations with geologic events in the northern Fennoscandian shield by dating them, using isotopic age determinations and by comparing pole positions with poles of previous studies from Fennoscandia.

2. General geology

The Tsuomasvarri intrusions are located in the northeastern part of the Fennoscandian shield which can be divided into four tectonic provinces (Gaál et al., 1989; Marker, 1990) (Fig. 2):

(1) The Palaeoproterozoic Lapland Granulite belt, which is composed of highly metamorphosed metaturbidites and lenses of intermediate to ultramafic igneous rocks. The age of igneous activity and metamorphism has been determined by isotopic age dating (U–Pb on zircon and titanite, Rb–Sr whole-rock method) to be 2000–1900 Ma old (Bibikova et al., 1973, 1991; Meriläinen, 1976; Bernard-Griffiths et al., 1984; Zykov et al., 1984; Huhma and Meriläinen, 1991). The Granulite belt is thrust southwest onto the Karelian Province.

(2) The Archaean Inari Terrain, which consists of three types of rocks: (i) light-coloured orthogneisses assigned ages of 2.73 and 2.5 Ga by U–Pb dating on zircons and reset ~1.9 Ga ago according to Rb–Sr dates (Meriläinen, 1976); (ii) supracrustal rocks consisting of amphibolites and migmatized metasediments and regarded as greenstone belts of Archaean (Meriläinen 1976) or Proterozoic (Bugge, 1980) age; (iii) large bodies of Hetta-type tonalites and granodiorites in the northern part of the Inari Terrain (Kesola, 1992). Some of these are associated with ultramafic rocks, such as the Tsuomasvarri body. U–Pb (zircon) ages of tonalites and granodiorites, which can be related to the quartz diorite–gabbro intrusions, are 1.95–1.90 Ga (Meriläinen, 1976). Kesola (1991) regards this as the age of metamorphism through which the Hetta-type granitoids were formed from Archaean gneisses. However, the U–Pb (zircon) age of 1.93 Ga, for the Tsuomasvarri gabbro-diorite (M. Vaasjoki, in Forsberg-Heikkilä, 1987) has been considered as the age of emplacement of the intrusion.

(3) The rift-related Opukasjärvi, Surnujärvi and Pechenga greenstone belts (i.e., the Kola suture belt of Berthelsen and Marker, 1986; Fig. 2) are mostly composed of low- to medium-grade metamorphic mafic metavolcanites and metasediments (Hanski and Smolkin, 1989; Hanski et al., 1990; Kesola, 1991). According to Kesola (1991), the Opukasjärvi greenstone belt is lithologically similar to the greenstone belt in the western part of the Inari Terrain and is Late Archaean or Palaeoproterozoic in age. So far, however, the isotopic age determinations (K–Ar on biotite) on the Opukasjärvi greenstone belt give only metamorphic ages of 1.81–1.70 Ga (Zwicky, 1991).

The Pechenga metavolcanic rocks in the Rus-

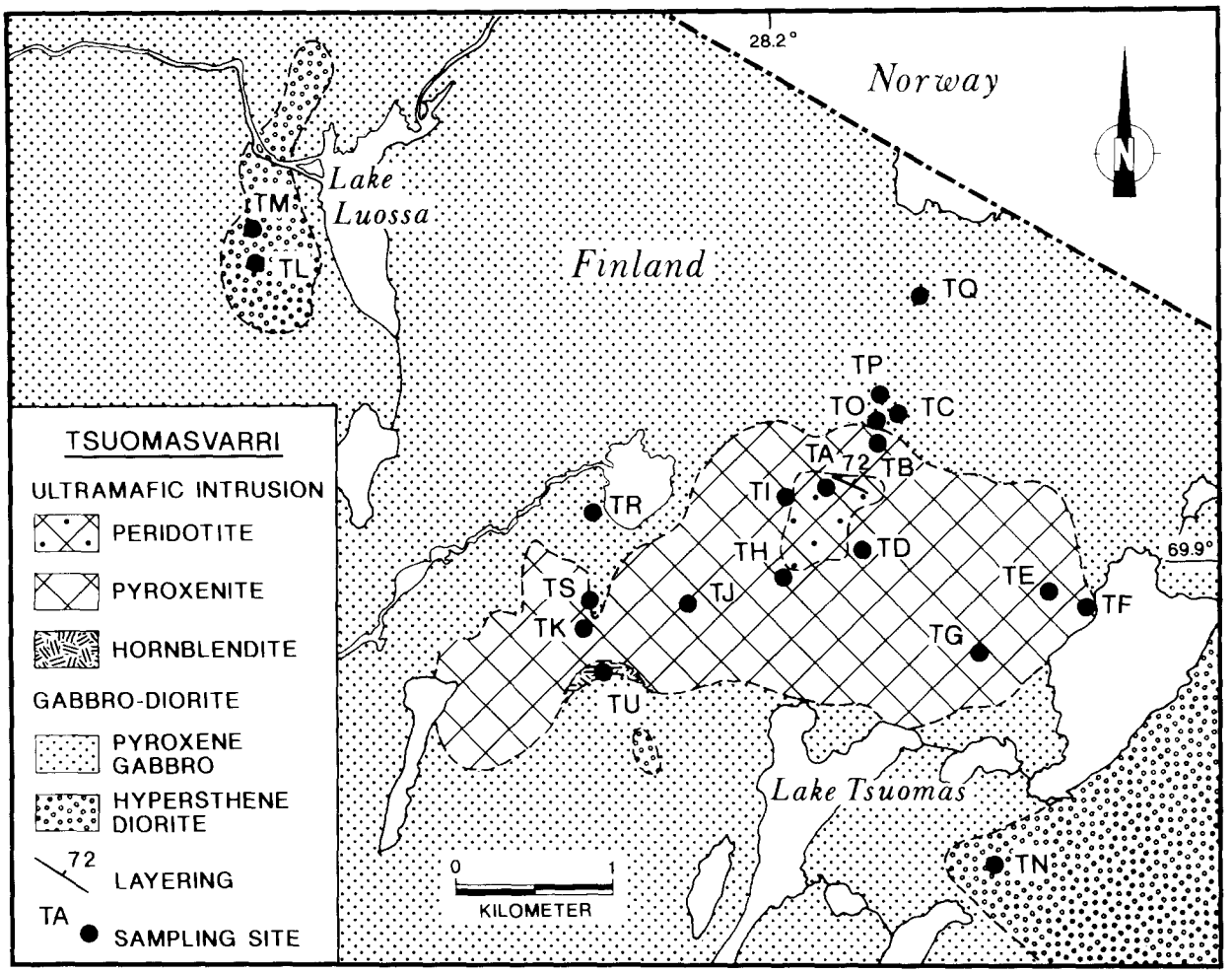


Fig. 3. Geological map of the Tsuomasvarri area (after Forsberg-Heikkilä, 1989)

sian part of the belt differ in chemistry and age from the Opukasjärvi metavolcanites (Kesola, 1991). According to Hanski et al. (1990) the age of the Pechenga picrites is 1.97–2.0 Ga. In the northern part of the Pechenga belt and its continuation to the southeast, the Varzuga belt, there are several basic layered intrusions (Fig. 1), such as Mt. Generalskaya of the age of ~2453 Ma (Sm–Nd determination; see e.g. Hanski et al., 1990), Moncha and Panski-Fedorova of the age of 2490–2453 Ma (Mitrofanov et al., 1991). These intrusions are only slightly older than the layered intrusions of Koillismaa in Finland and in northern Karelia in Russia (ages 2435–2445 Ma, Alapieti, 1982; Krasnova and Gooskova, 1990). The Opukasjärvi and Surnujärvi greenstone belts are separated by the large Vainospää granite (U–Pb age 1790 Ma; Huhma, 1986), emplaced in the post-orogenic stage of the Svecokarelian orogeny.

(4) The geochemical and geophysical features of the Archaean Sörvaranger Terrain, differ from the Archaean Inari Terrain (see Marker, 1990). The Sörvaranger Terrain consists of two NW–SE-trending supracrustal belts surrounded by orthogneisses and migmatites. The oldest intrusions in the Sörvaranger Terrain are ~2.55 Ga old (U–Pb zircon age; Meriläinen, 1976) and the

youngest 1673 ± 165 Ma old (Rb–Sr dating; Krill, 1985).

3. Geology of the Tsuomasvarri intrusions

3.1. Ultramafic central intrusion

The Tsuomasvarri ultramafic body (Fig. 3), covering an area of about 8 km^2, is a cone-like layered intrusion surrounded by a gabbro-diorite intrusion (Papunen, 1977; Forsberg-Heikkilä, 1989). The main rock type in the ultramafic body is olivine pyroxenite and the others are serpentinite dunite, serpentinite, peridotite and hornblendite (Forsberg-Heikkilä, 1989). In some areas the rhythmical alternation of pyroxenite and dunite layers is well preserved. The dip of layering is steep, about 70°, towards north-northeast (Fig. 3). The dip of layering is primary, formed during the emplacement of the ultramafic magma and has not changed in any later deformation (R. Kesola, pers. commun., 1986).

The rocks are generally well preserved, but in the northern part of the intrusion, the dunites are strongly serpentinized (Forsberg-Heikkilä, 1989). Serpentinite dunites are often cut by thin magnetite veins. In the pyroxenites, primary magnetite occurs mostly as small inclusions in augite grains and as a deuteric alteration product of olivine and pyroxenes.

3.2. Gabbro-diorite

In the gabbro-diorite intrusion, rock types vary from pyroxene gabbros to pyroxene diorites (Forsberg-Heikkilä, 1989). Based on varying compositions within the intrusion, Kesola (1992) has suggested that the gabbro-diorite was formed through ultrametamorphism and remobilization of the originally Archaean gneisses. This concept is supported by relics of gneisses within the gabbro-diorite (for example, site TQ, Fig. 3) and gradual change of the gabbro-diorite into the surrounding Archaean gneisses.

The most typical rock type is a partly hypidiomorphic pyroxene–hornblende gabbro. In thin sections, magnetite is visible as small inclusions in pyroxenes and biotite and as an alteration product of pyroxenes and hornblende (see also Forsberg-Heikkilä, 1989). Another rock type is hypersthene diorite, with magnetite as inclusions in pyroxene and as an alteration product of pyroxenes.

3.3. Contacts between the ultramafic and gabbro-diorite intrusions

Contacts between the ultramafic body and the gabbro-diorite are exposed in the western (site TS) and southwestern (site TU) sites (Fig. 3). Forsberg-Heikkilä (1989) concludes that the central ultramafic intrusion is clearly younger than the surrounding gabbro-diorite intrusion for the following reasons:

(1) In the west, the contact between the gabbro-diorite and the ultramafic rocks is sharp and in the contact zone the gabbro is metamorphically altered as revealed by the strong saussuritization of plagioclase and the formation of secondary epidote. The amount of hornblende in the surrounding gabbro increases in the vicinity of the contact. Foliation in the gabbro is intense at the contact and it conforms to the shape of the ultramafic body.

(2) In the southwest, there are angular fragments of gabbro in a matrix of ultramafic hornblendite, indicating that the gabbro-diorite must have been, at least partly, in a solid state during the emplacement of the ultramafic body. The contact between the ultramafic intrusion and the gabbro, however, is not sharp, and in places the ultramafic hornblendite grades into gabbro as the amount of plagioclase in the hornblendite increases.

3.4. Isotopic age determinations

The gabbro-diorite intrusion has been dated by the U–Pb zircon method (M. Vaasjoki, in Forsberg-Hekkilä, 1989). Age determination was done from a gabbro pegmatite (site TL, Fig. 3), located about 2 km from the central ultramafic intrusion. According to M. Vaasjoki (op. cit.) the zircon occurs as crushed fragments of larger grains in two variants of different colour: trans-

parent dark brown and opaque light grey. Four analysed fractions define a discordia line with the upper intercept of the concordia curve at 1931 ± 2 Ma. This age is considered to be the age of crystallization of the gabbro-diorite intrusion. The lower intersection point in the concordia diagram gives an age of 399 ± 1 Ma. Radiometric ages are not yet available from the Tsuomasvarri ultramafic intrusion.

4. Sampling and laboratory procedure

The Tsuomasvarri intrusions form a well-exposed 435 m high hill. Fifty-five oriented core samples were taken from ten sites in the ultramafic intrusion (Fig. 3), using the techniques described previously in Mertanen et al. (1987). The majority of the samples (45) are olivine pyroxenites, six are serpentinite dunites and four are hornblendites (see Fig. 3). In the surrounding gabbro-diorite 46 samples were taken from ten sites (Fig. 3).

The NRM was measured on spinner magnetometers (Pesonen et al., 1983). AF demagnetizations were done with a single-axis demagnetizer (in steps of 10 mT, up to 100 mT), and thermal demagnetization was performed (in steps of 100°C up to 500°C and then in 20–10°C steps up to 680°C) in Schonstedt or homemade furnaces (Leino, 1979; Pesonen et al., 1983). Susceptibilities were measured in the course of thermal demagnetizations (in steps of 200°C, 300°C, 400°C, 500°C, 520°C, 570°C, 600°C, 640°C and 680°C) to detect mineralogical changes, interpreted on the basis of changes in bulk susceptibility during heating. Magnetizations were isolated using principal component analyses (Kirschvink, 1980) with maximum angular deviation (MAD) values of 6° in most samples and up to 10° in some cases with larger scatter. Stereographic projections, demagnetization decay curves and orthogonal vector plots (Zijderveld, 1967; Torsvik, 1986) were used for examination of the response of specimens to demagnetization and for visual inspection of straight-line segments, respectively.

Hysteresis analyses on ten samples were done at the Technical Research Centre of Finland. Petrophysical properties, such as density, susceptibility and Koenigsberger Q-ratio, were determined on each specimen. Polished thin sections were examined for samples carrying well defined magnetization components.

5. Palaeomagnetic results

In the beginning of our study, we had good pilot samples showing hard and stable magnetizations. In those samples one or two remanent magnetization components were clearly isolated. However, in the course of investigations, it became apparent that the rocks have been remagnetized or partially overprinted in at least two or three geological events. Probably due to the differing abilities of the rocks to resist remagnetizations, some magnetizations are encountered only sporadically throughout the various sites. Usually one specimen carries one or two different components and the components vary from specimen to specimen. In addition, most samples reveal a large component of the present Earth's field (PEF) magnetization which also diminishes the number of accepted samples. For this reason the number of samples used in the analysis is unfortunately fairly low for obtaining statistically well defined means. However, we report the results of these investigations, knowing that the interpretations are based on limited data.

For the Tsuomasvarri ultramafic intrusion, we have separated five remanence components (hereafter called A, B, C, D and F) all of which we believe to have geological significance. The ultramafic intrusion is dominated by a well-grouped southeasterly and moderate upward magnetization, C. Most samples carry multi-component magnetization with two to three superimposed remanence components. The other components are statistically less well grouped, probably owing to the small number of samples having a particular component.

The gabbro-diorite is dominated by a component called A′, the direction of which is nearly antipodal to the C direction in the ultramafic body. In addition, magnetization D, equivalent

to D in the ultramafic body, is separated from some sites. In the next we describe each remanence component. The formation mean magnetization directions of all components and corresponding palaeomagnetic pole positions are given in Tables 1 and 2 and in Fig. 11. The site mean results for each component are given in Appendices 1 and 2.

5.1. Ultramafic intrusion

The characteristic magnetization C in the Tsuomasvarri ultramafic intrusion has a southeasterly declination and shallow negative inclination ($D=146.9°$, $I=-17.9°$, $\alpha_{95}=8.4°$, $k=33.9$, $N=10$ sites, $n=32$ samples). Magnetization C was isolated at all ten sites in the body (Table 1 and Appendix 1). Fig. 4 shows a typical behaviour of a specimen (TI41-1A) carrying C, revealed as a stable end point after first removing another component, designed as F. The discrete unblocking temperature spectra of specimens mainly revealing (in addition to the softer PEF component) the C component (Fig. 5, specimen TE21-1E) is interpreted to show that the C magnetization is carried by low Ti-magnetite with laboratory unblocking temperatures (T_{LUB}'s) slightly below 600°C.

The northeasterly and moderate downward direction of magnetization designed as A ($D=341.1°$, $I=43.5°$, $\alpha_{95}=5.0°$, $k=182.8$, $N=6$ sites, $n=9$ samples) is almost antipodal to the characteristic component C. However, when the A directions are switched by 180°, the cones of 95% confidence around the mean A and mean C directions do not intersect, suggesting that they are significantly different and were acquired at slightly different times. Difference between the inclinations of A and C (irrespective of sign) is about 25° and that of the declinations about 15°. A has been isolated in the pyroxenite sites near the contact with the surrounding gabbro-diorite, but not in the serpentinite dunites in the middle of the intrusion.

Components A and C co-exist in three samples (eight specimens) in which A is the low-coercivity component during AF demagnetization. In Fig. 6 (specimen TF27-1A), the lower-intensity component A was isolated in the coercivity range 0–10 mT. However, since the A differs clearly from the PEF magnetization, but is close to the typical magnetization acquired during the Svecokarelian orogeny ~1.9–1.8 Ga ago (e.g. Pesonen et al., 1991), we interpret that A is not a recent viscous remanent magnetization. The C component with higher intensity is erased in the range 15–100 mT. The straight line revealing C (Fig. 6) runs towards the origin, suggesting an absence of other high-coercivity magnetizations.

The behaviour of the samples in thermal demagnetization is more complicated. In some thermally demagnetized specimens, A was iso-

Table 1
Grand mean magnetization directions and pole positions of the Tsuomasvarri ultramafic intrusion (69.9°N, 28.3°E)

Component	N	Mean remanence direction				Pole position					
		D (°)	I (°)	α_{95} (°)	k	Plon (°N)	Plat (°E)	$A95$ (°)	dp (°)	dm (°)	g
A	6	341.1	43.5	5.0	182.8	44.1	232.4	4.9	3.9	6.2	A
B	5	42.7	38.2	13.0	35.9	35.4	157.6	10.7	9.1	15.3	B
C	10	146.9	−17.9	8.4	33.9	25.7	245.1	7.0	4.5	8.7	B
D	6	93.7	47.0	14.6	21.9	−25.1	284.5	15.0	12.2	18.9	C
F	5	176.3	31.9	15.7	24.6	2.7	211.8	16.2	10.0	17.7	C

N=number of sites used to calculate the mean magnetization direction and pole; D=declination; I=inclination; α_{95}=the radius of a cone of 95% confidence about the mean; k=the precision parameter (Fisher, 1953); A_{95}=the radius of a cone of 95% confidence about the mean poles determined from the site mean D and I; dp, dm=the semiaxes of an oval of 95% confidence about the poles; g=modified Briden-Duff (Briden and Duff, 1981; Pesonen et al., 1989) reliability grade ranging from A (most reliable) to D (least reliable).

Table 2
Grand mean magnetization directions and pole positions of the Tsuomasvarri gabbro-diorite intrusion (69.9°N, 28.3°E)

Component	N	Mean remanence direction				Pole position					
		D (°)	I (°)	α_{95} (°)	k	Plat (°N)	Plon (°E)	A_{95} (°)	dp (°)	dm (°)	g
A′	8	328.2	41.4	6.3	79.0	40.2	247.3	6.8	4.7	7.7	A
D	3	95.3	40.2	19.0	43.2	−19.7	285.2	17.4	13.8	22.9	C

For explanations, see Table 1.

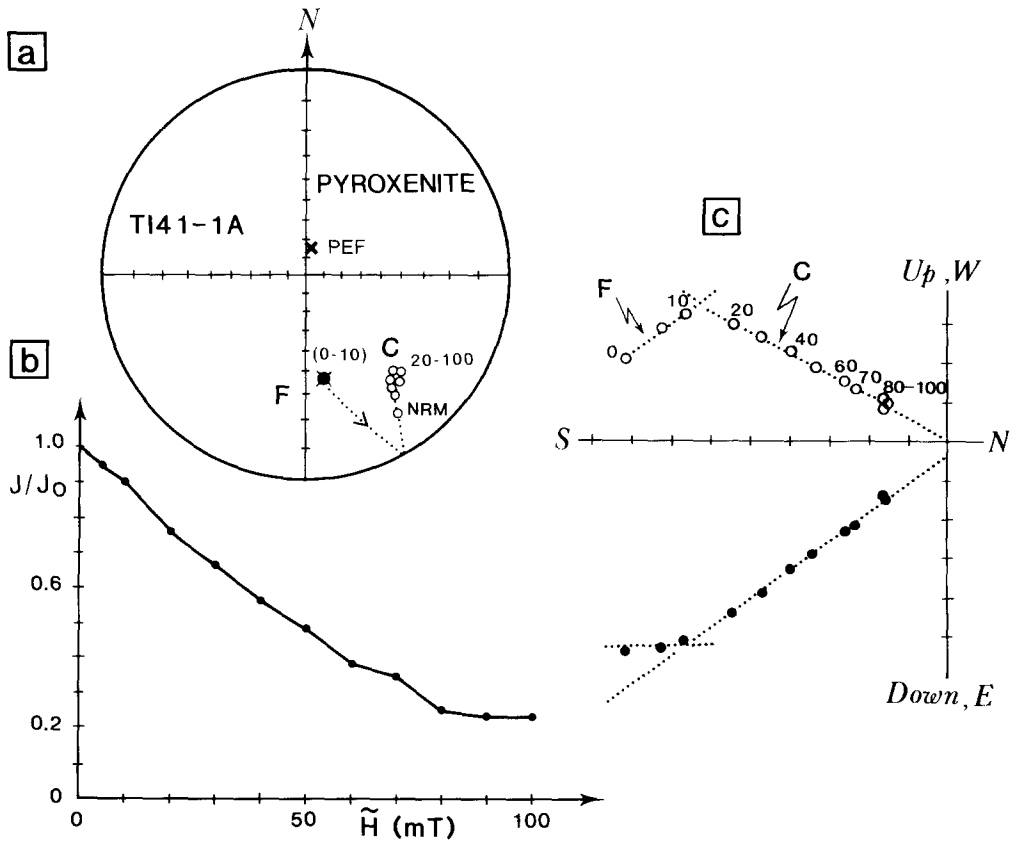


Fig. 4. AF demagnetization data of specimen TI41-1A. (a) Equal-area stereoplot, with solid (open) symbols denoting downward (upward) projections. Cross with circle denotes the subtracted vector over the field range given in parentheses. (b) Normalized intensity decay curve. (c) Orthogonal component plot with solid (open) symbols indicating the data plotted onto the horizontal (vertical) plane. PEF (×) is the present Earth's field at the site locality. Specimen TI41-A reveals a typical C component after removal of the softer F component.

lated at lower laboratory unblocking temperatures than C but in some A displays higher T_{LUB}'s than C. In some specimens A was isolated at both low (200–350°C) and high (580–640°C) T_{LUB}'s while C was isolated at intermediate T_{LUB}'s (500–580°C). Such behaviour can be explained in two ways. First, A may represent a thermochemical remanent magnetization (TCRM) acquired at moderate temperatures, but the remanence has blocked in grains which will have

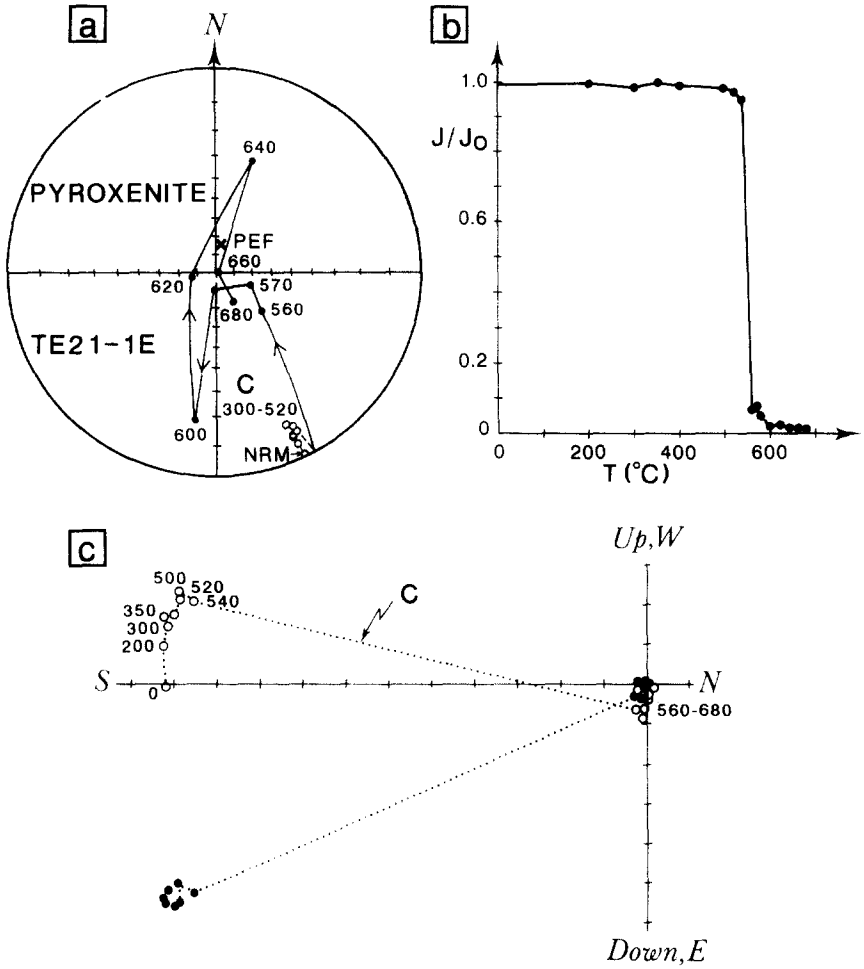


Fig. 5. The discrete laboratory unblocking temperature demagnetization decay curve of specimen TE21-1E indicates that the remanence is carried by fine low Ti-magnetite. Symbols as in Fig. 4.

both high T_{LUB}'s (chemically formed magnetite grains) and low T_{LUB}'s (thermally blocked grains). Second, low and high blocking temperatures of the same remanence component are possible if the high (580–640°C) T_{LUB} component is carried by hematite (Curie point 680°C) and the low (200–350°C) T_{LUB} component by coarse-grained titanomagnetite.

Component B (D=42.7°, I=38.2°, α_{95}=13.0°, k=35.9, N=5 sites, n=7 samples, Table 1 and Appendix 1) is not well grouped. It was isolated at five sites in different parts of the intrusion. At four sites it was isolated in only a single sample. The T_{LUB} range of B (560–600°C) indicates that B is carried by Ti-poor magnetite. For example, in specimen TS45-1C (Fig. 7), component B was isolated in a temperature range of 560–600°C after removal of the PEF magnetization. In two samples B co-exists with C and in one sample with D. In both cases component B is the high T_{LUB} component. In a sample subjected to AF demagnetization, B was isolated at lower coercivities than C, but in other samples B was isolated (after the viscous component) over a range of high peak inductions.

Component D (D=93.7°, I=47.0°, α_{95}=14.6°, k=21.9, N=6 sites, n=11 samples, Table 1 and Appendix 1) was isolated from the

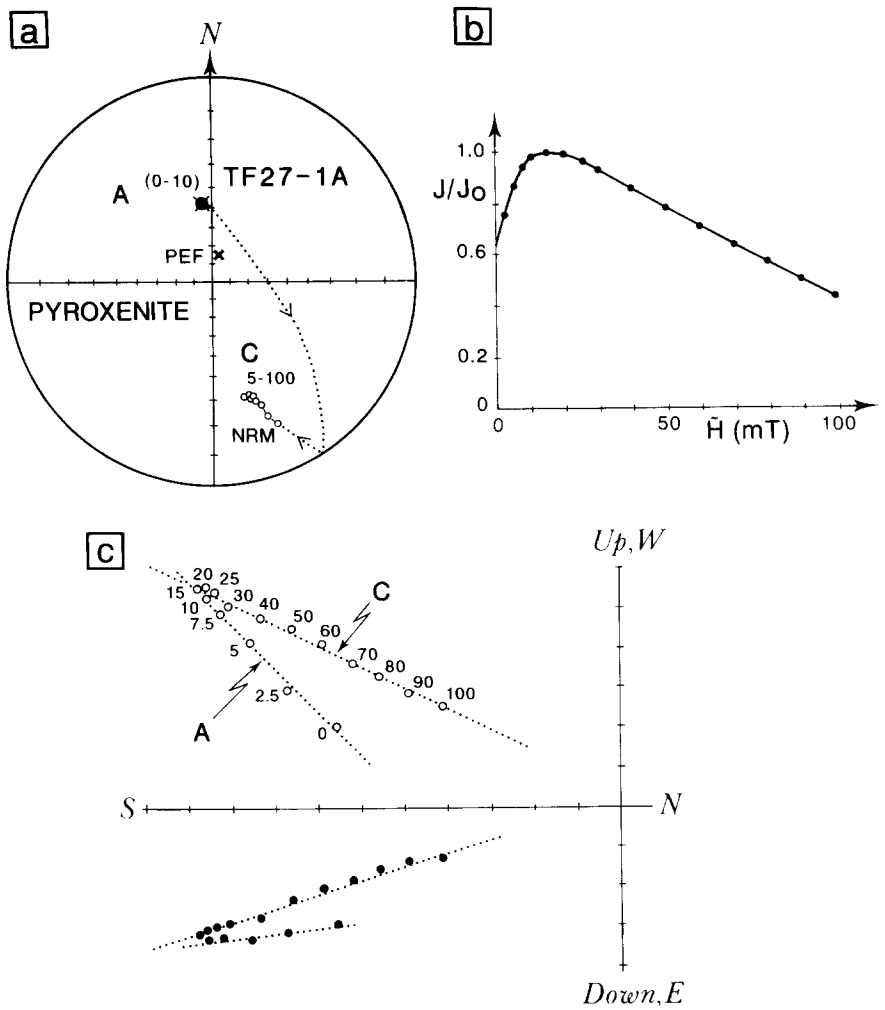


Fig. 6. In AF demagnetized specimen TF27-1A the normal polarity component A is revealed only in the orthogonal plot. The reversed polarity component C is isolated in the orthogonal plot as a well defined linear trajectory. Symbols as in Fig. 4.

serpentinized dunites and pyroxenites. In two samples D is the only component revealed (besides PEF). In all others D co-exists with other components. In most samples D co-exists with C and shows a range of high T_{LUB}'s (400 to 680°C). In AF demagnetized samples, D is the low-coercivity component, as for example in specimen TA5-1F (Fig. 8), where D co-exists with PEF. Above 40 mT there is no decline in intensity to testify that a high-coercivity component is demagnetized. In an initial report on the Tsuomasvarri ultramafic intrusion (Mertanen and Pesonen, 1992), component D was interpreted to be a composite of A and C. However, our new data on the gabbro-diorite intrusion refute this hypothesis, because component C does not exist in the gabbro-diorite whereas component D does.

Component F ($D=176.3°$, $I=31.9°$, $\alpha_{95}=15.7°$, $k=24.6$, $N=5$ sites, $n=5$ samples, Table 1 and Appendix 1) was isolated at sites in the north and east of the intrusion. Component F is of low T_{LUB} and low coercivity. In most samples magnetization F co-exists with component C (see Fig. 4), and in one sample with component D. In both cases it is the first component to be isolated. The T_{LUB}'s (typically below 560°)

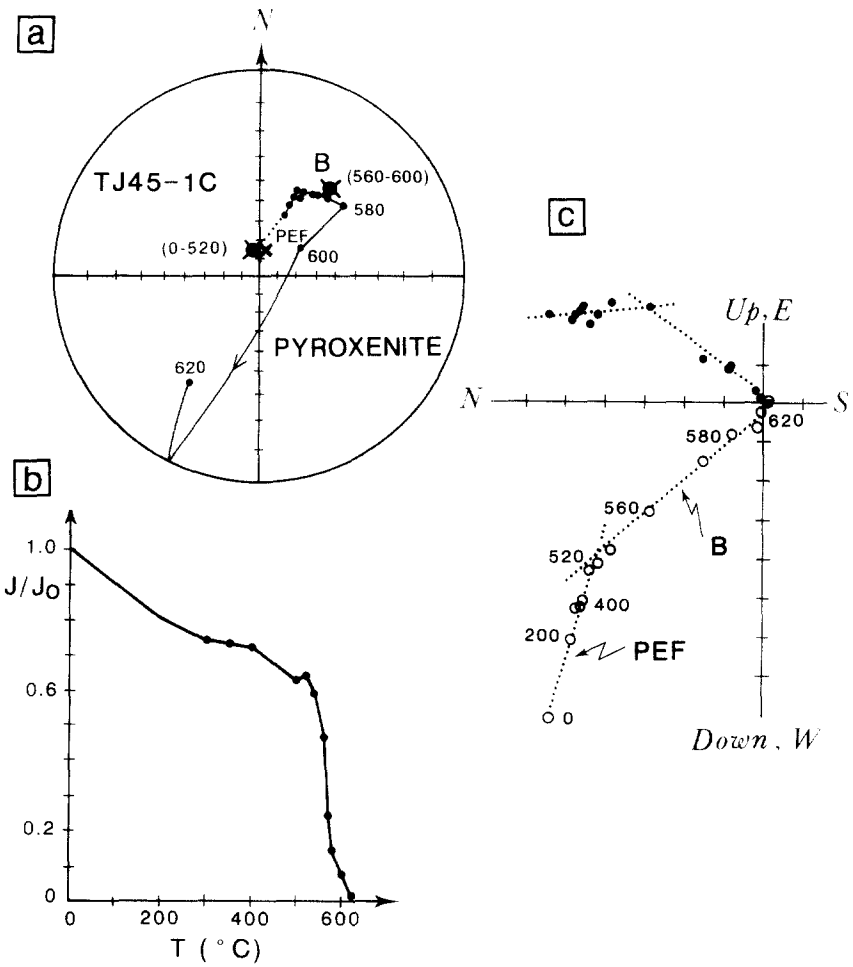


Fig. 7. In sample TJ45-1C remanence component B is isolated after removal of the viscous PEF component. For conventions, see Fig. 4.

of F suggest that it was acquired at relatively lower temperatures than the other components and that it is carried by titanomagnetite or by pyrrhotite.

5.2. Gabbro-diorite intrusion

The gabbro-diorite intrusion is dominated by component A′ ($D=328.2°$, $I=41.4°$, $\alpha_{95}=6.3°$, $k=79.0$, $N=8$ sites, $n=28$ samples, Table 2 and Appendix 2) which differs slightly, but significantly (the 95% confidence cones do not intersect), from the A magnetization of the ultramafic intrusion (Fig. 11). The discrete T_{LUB} (after removal of the softer PEF component from all samples) and maximum unblocking temperatures around 580°C of most of the samples indicate that A′ is carried by low-Ti titanomagnetite. We interpret A′ to be of thermal origin (TRM).

At sites TC, TO, TP and TS, located within 175 m of the contact to the ultramafic body, the NRM intensities are so weak that stable results could be obtained only from one or two samples per site. The direction of A′ at site TS is statistically distinct from all other site mean directions (Table 2 and Appendix 2) and may be attributed to

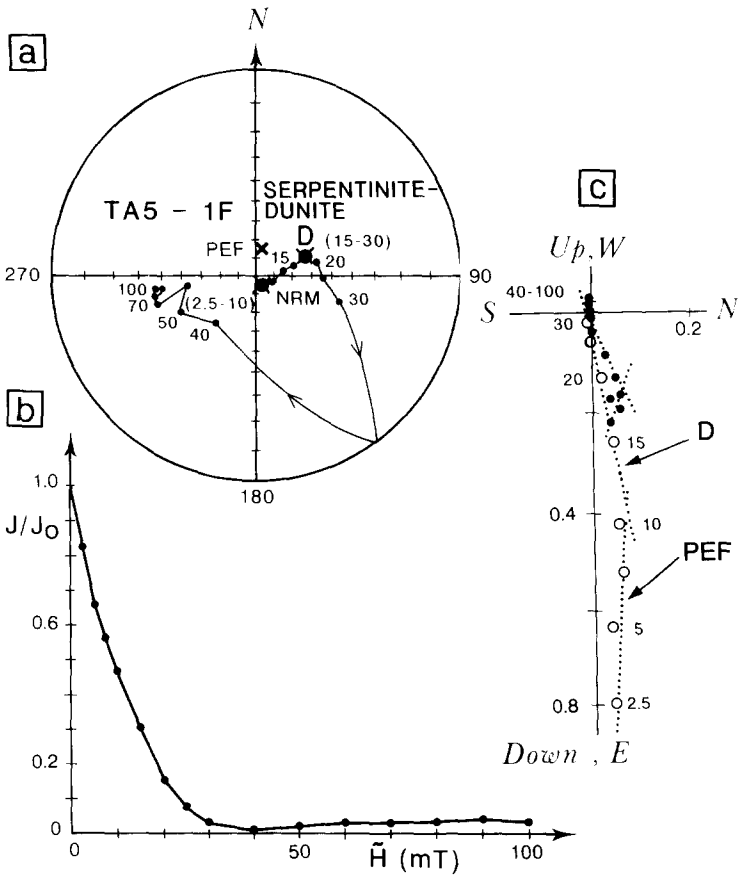


Fig. 8. Specimen TA5-1F carries the soft PEF component and the intermediate D component after which there is no decrease in the intensity decay curve showing an existence of any other stable magnetization. For conventions, see Fig. 4.

its location in a wedge, where the gabbro is strongly foliated along the contact with the ultramafic body (Fig. 3). The divergent A′ direction at this site probably indicates that the blocking of remanence occurred shortly before the deformation and hence, the ultramafic intrusion postdates the gabbro-diorite, which was in an almost solid state at the time of the emplacement of the ultramafic body.

At site TQ (Fig. 3), about 750 m from the contact, the ultramafic intrusion has not, however, deformed the gabbro-diorite. The latter is foliated and stratified and bears a close resemblance to the surrounding Archaean gneisses. The direction of A′ at this site corresponds to the overall mean A′ direction of the other gabbro-diorite sites (Appendix 2), indicating that A′ post-dates the foliation.

In one specimen (TS1-1B, Fig. 9), located 1 m from the contact with the ultramafic intrusion, we have isolated a magnetization which corresponds to the previously described C component of the ultramafic body. This provides, albeit meagre, evidence of the baking effect of the ultramafic magma. If so, the C component in the ultramafic body would record a primary intrusion magnetization. Unfortunately, the remanence intensity is too weak in most other contact gabbro-diorite samples to yield meaningful results.

At three gabbro-diorite sites, TM, TN and TO, there occurs another remanence component

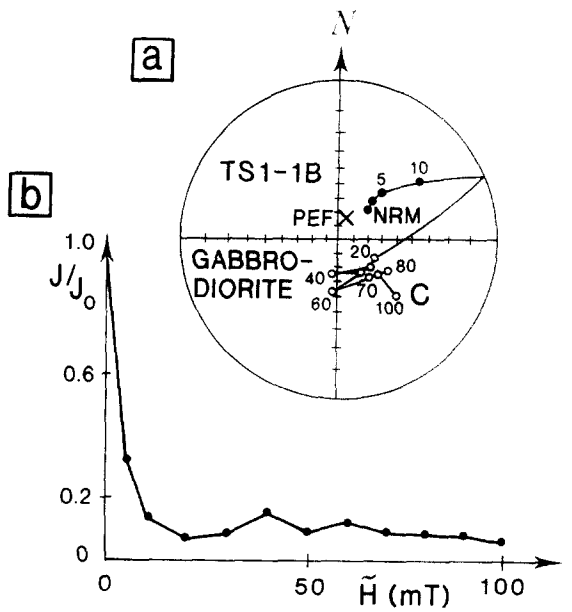


Fig. 9. Specimen TS1-1B from the gabbro-diorite carries a large PEF component and a magnetization corresponding to component C in the ultramafic intrusion. For conventions, see Fig. 4.

which is not precisely isolated. The direction is within the same quadrant ($D=95.3°$, $I=40.2°$, $\alpha_{95}=19.0°$, $k=43.2$, $N=3$ sites, Table 2 and Appendix 2) as the D direction in the ultramafic intrusion. In all gabbro-diorite samples, D was isolated as a high T_{LUB} component. However, as is evident from specimen TN2-1A (Fig. 10), the directions at high temperatures are quite scattered, although the intensity is well above the noise level. At site TO, D was isolated in one AF demagnetized specimen in the field range 0–5 mT, but even here D is distinct from PEF.

6. Petrophysical and petrological properties of the rocks

Alternating field and thermal demagnetizations isolated five magnetizations (A, B, C, D and F) in the ultramafic intrusion and two (A′ and D) in the gabbro-diorite intrusion (Fig. 11). However, noting the small number of samples in which the components were isolated, their significance and relative ages are difficult to interpret. We have also studied the petrophysical and petrological properties of the rocks in order to seek correlations that might help us to resolve the origin of each component and infer the ages of remanence acquisition. The petrophysical relation diagrams are shown in Fig. 12. Hysteresis measurements provided information on domain structure and magnetic grain sizes in ten samples with different magnetizations.

6.1. Ultramafic intrusion

The characteristic NRM intensities of the serpentinite dunites and pyroxenites vary from 300 to 11,800 mA m^{-1} and the susceptibilities from 2300 to $57{,}000\times10^{-6}$ SI. The corresponding Koenigsberger Q-values range from 1 to 10 (Fig. 12).

The lowest densities and highest susceptibilities and NRM intensities are in the *serpentinite dunites* (Fig. 12), where the D magnetization is dominant. The high susceptibility and remanence intensity values probably reflect fine-grained magnetite. Forsberg-Heikkilä (1989) describes magnetite as a main mineral in fine grained serpentinite dunites, with abundances of up to 11% (modal composition). In thin sections a considerable proportion of magnetite is a high-temperature deuteric alteration product of olivine. As implied by the skeletal structure of olivine, serpentinization and the associated formation of magnetite took place during late stages of the primary cooling of the ultramafic magma (Forsberg-Heikkilä, 1989).

In addition to the primary magnetite, numerous magnetite veins (ca. 2 mm wide) probably carry secondary magnetization. The intermediate Q-values of the serpentinite dunites, about 2 to 4 (Fig. 12b), indicate that the carriers of magnetization are SD (single domain) or PSD (pseudo-single domain) grains (e.g. Collinson, 1983). Hysteresis measurements support this interpretation and show that the magnetization is carried by PSD grains (Fig. 13) (Day et al., 1977). However, since these samples have clearly low NRM coercivities and clear signs of strong viscous PEF contaminations, it is obvious that the PSD grain-size characteristics reflect the su-

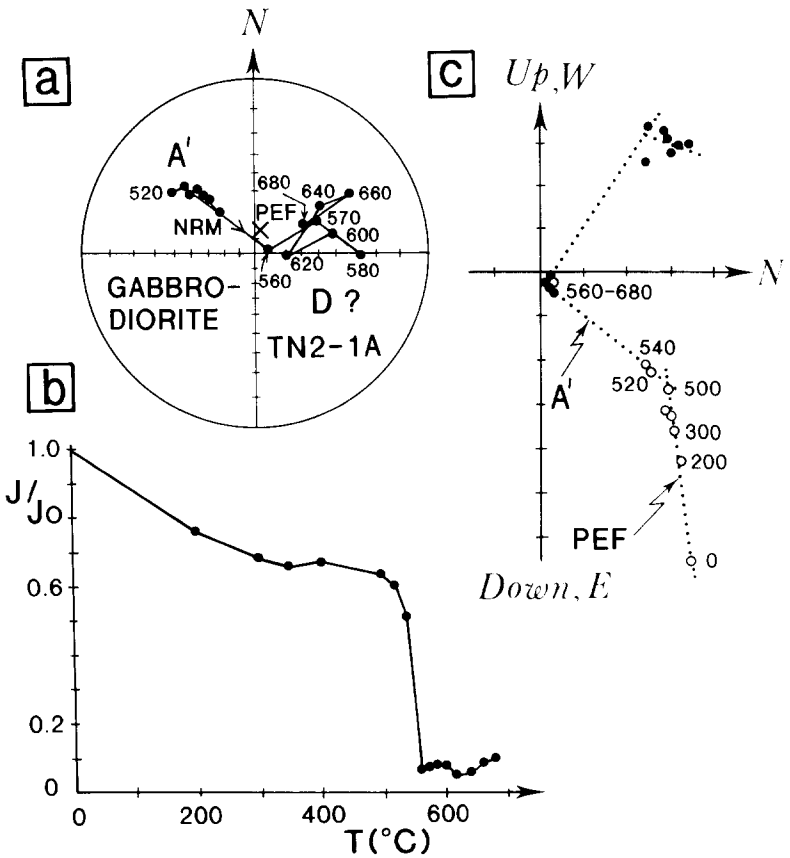


Fig. 10. In specimen TN2-1A from the gabbro-diorite, the occurrence of PEF and A magnetizations are shown in the orthogonal vector plots. In high temperatures the data are scattered, but the direction is within the same sector as magnetization D in the ultramafic intrusion. For conventions, see Fig. 4.

perposition of a small fraction of SD grains and a large fraction of multi-domain (MD) magnetite grains.

The density of *pyroxenites* (Fig. 12a) tends to increase with the decrease of susceptibility. According to Puranen (1989), high densities of most igneous rocks reflect high iron contents, which favour magnetite formation. However, the modal compositions of pyroxenites (Forsberg-Heikkilä, 1989) show that the abundance of magnetite is only about 1%. Thus the high densities are caused by iron incorporated in the silicate mineral structures.

According to microscopic studies, two types of magnetite occur in pyroxenites. First, magnetite occurs as very fine, disseminated grains in augite crystals. Second, magnetite also occurs as partly hypidiomorphic (grain size about 0.1 mm) grains between pyroxene grains. The comparably high Q-values of 1 to 10 (Fig. 12b) indicate that the magnetization is carried by SD or PSD grains. The hysteresis results of all pyroxenites (Fig. 13) support this (Day et al., 1977). Because the threshold size between MD and PSD grains for magnetite is about 15–40 μm (Day, 1977; Dunlop, 1983), all microscopically visible magnetite is of MD grain size. Thus, all samples carrying multicomponent magnetizations with PSD hysteresis properties are in fact mixtures of MD and SD/PSD grains.

The occurrence of a high-coercivity magnetite is evident from the shape of the thermal demag-

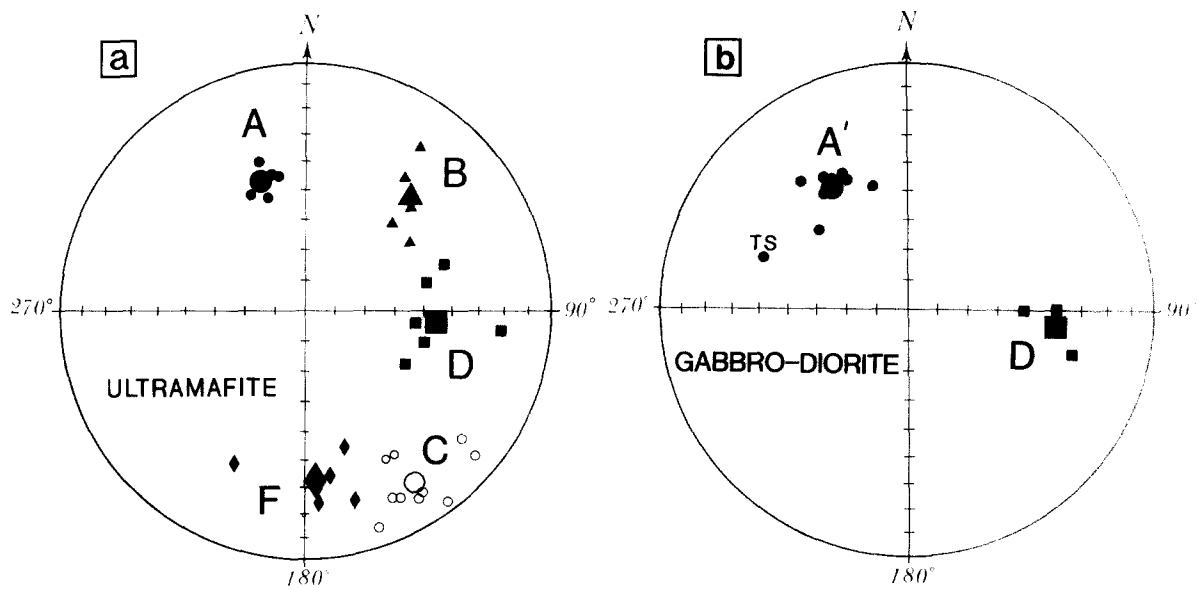


Fig. 11. Equal-area projection of magnetization directions from the Tsuomasvarri ultramafic intrusion (a) and gabbro-diorite (b). Small symbols are site means, larger symbols are grand means for each magnetization. Closed (open) symbols denote normal (reversed) polarity; dots represent component A (ultramafic rocks) or A′ (gabbro-diorite); triangles, component B; circles, component C; squares, component D; diamonds, component F. See also Tables 1 and 2 and Appendices 1 and 2.

netization curves and from the T_{LUB}'s of about 570°C (for example, specimen TJ21-1E, Fig. 5). In this specimen the only component isolated is C and it may thus be a primary magnetization. Component C also occurs in specimens TE23-1F, TF27-1A and TI41-1A (Fig. 13), but now together with components D, A and F, respectively. In all these specimens magnetite occurs in at least two phases.

6.2. Gabbro-diorite intrusions

All petrophysical values are lower in the gabbro-diorite than in the ultramafics: NRM intensities range between 0.1 and 1000 mA m^{-1} and susceptibilities between 640 and 41,000 x 10^{-6} SI. Most Q-values of the gabbro-diorite are in the range 0.01–2.5.

The lowest susceptibilities and remanences with highest densities (Fig. 12) are from gabbro-diorite sites TO, TP and TU, close to the contact with the ultramafic body. Most samples of these sites are composed of coarse silicate minerals with only a minor amount of magnetite and no measurable remanence. On the contrary, the highest susceptibility and remanence values and the lowest densities are carried by hypersthene diorite samples from sites TL, TM, and TN, where the most stable remanence directions were also isolated. Samples from site TQ exhibit the greatest variation in susceptibility and remanence, obviously due to the large variation in mineral composition.

Hysteresis measurements (Fig. 13) show MD characteristics for the gabbro-diorite samples. This is evidently due to the large MD grains (diameter up to 1 mm) carrying the PEF component of viscous origin which dominate the PSD/SD grains carrying the stable A′ component.

The variation in the petrophysical properties of the gabbro-diorites, together with the differences in the remanence directions at the sites, implies that the composition of the original magma source is very variable and that, near the contact, the ultramafic body has baked the gabbro-diorite, thus leading to a decrease in remanence intensity.

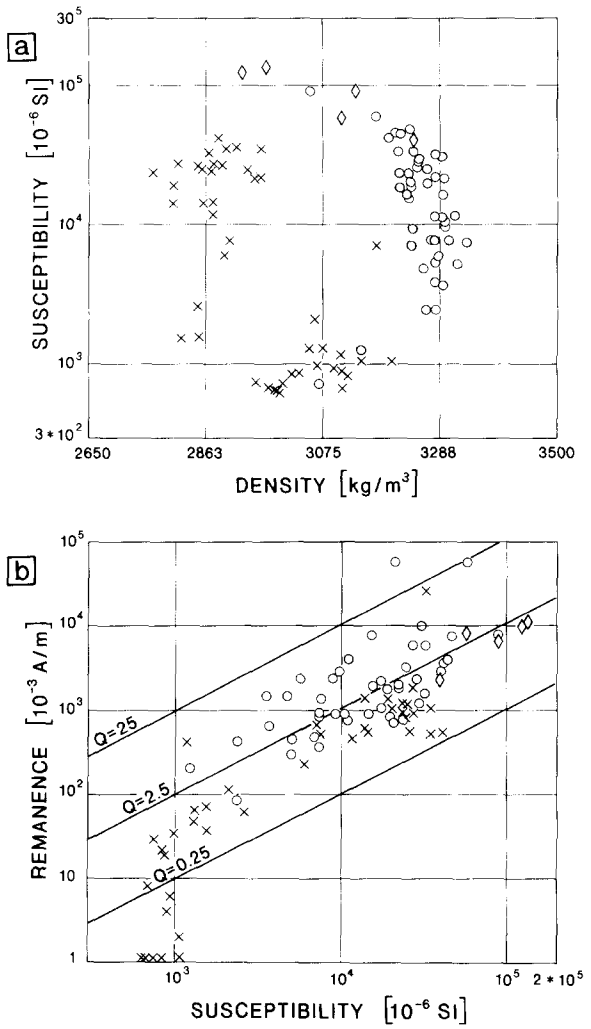


Fig. 12. (a) Susceptibility–density, (b) remanence–susceptibility relations of rocks in the Tsuomasvarri ultramafic and gabbro-diorite intrusions. Diamonds represent serpentinite dunite; circles, pyroxenite; crosses, gabbro-diorite.

7. Interpreted ages of magnetization

Magnetizations isolated in the Tsuomasvarri intrusions are plotted in Fig. 11. The primarity of one of the components could not be verified by tectonic tilt tests, since the attitudes of the layerings are original (R. Kesola, pers. commun., 1986). In addition, the relative ages of magnetizations are difficult to define because of overlapping blocking temperature or coercivity spectra. However, because many samples carry only two components, relative ages in some cases can be determined. We summarize the above palaeomagnetic, rock magnetic and petrological evidence for the following interpretation.

(1) Thermal demagnetization curves show that the characteristic component C of the ultramafic intrusion is of thermal origin. Petrophysical and hysteresis data imply that it is carried by SD/PSD magnetite. C is in all likelihood of primary origin, formed during cooling of the magma. The limited results from one baked gabbro-diorite specimen from the contact, also carrying C, support this interpretation.

(2) Component A is probably a thermochemical remanence (TCRM) carried by hematite or coarse-grained magnetite. As implied by its commonly broader range or low T_{LUB}'s it is probably slightly younger than C.

(3) B was separated only in a few samples; its relative age cannot be reliably determined from the T_{LUB} or coercivity spectra.

(4) In most of the samples in which both C (ultramafic body) or A′ (gabbro-diorite) and D were isolated, the latter is the high temperature T_{LUB} component. D (as well as C) was separated in most of the serpentinite dunite samples, in which the bulk of the magnetite is primary, formed in deuteric alteration during magmatic cooling. D could thus be primary. This magnetization is also found in the gabbro-diorite, close to the contact (site TO) and at sites TM and TN, where it occurs as a high T_{LUB} temperature component. However, the fact that in some samples the blocking temperatures of D and in most samples the coercivities of D are lower than those for C, support the secondary origin for D. Although D is close to PEF, they are still distinct, when isolated in the same specimens.

(5) F is the youngest magnetization in the ultramafic body. It is secondary, as shown by its clearly lowest T_{LUB}'s and coercivities.

(6) Component A′ is characteristic of the gabbro-diorite intrusion and is interpreted to be a primary TRM.

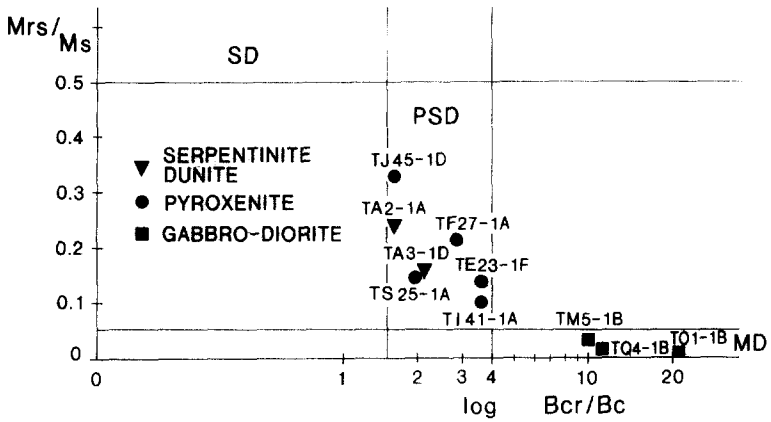


Fig. 13. Hysteresis properties of different rock types (diagram after Day et al., 1977).

8. Discussion

Palaeomagnetic poles were calculated for the Tsuomasvarri intrusions from the grand mean magnetization data for each component. The poles were plotted on the Precambrian (Fig. 14) and Palaeozoic (Fig. 15) APWPs of the Fennoscandian shield (Elming et al., 1993). Components A, A′, B, C and D have previously been reported from the Fennoscandian shield (see, e.g., Mertanen et al., 1989), but this is the first time that pole F has been defined in the Proterozoic rocks of Fennoscandia. Due to the small number of samples revealing magnetizations A, B, D and F, the poles are too poorly defined to provide more than an indication of the magnetization age of remanence. In the following, the ages of magnetization acquisition and geotectonic aspects of the northeastern Fennoscandian shield are examined in the context of palaeomagnetic poles.

8.1. Ultramafic intrusion

Pole C (with reversed polarity) (Fig. 14) is the characteristic magnetization in the Tsuomasvarri ultramafic intrusion and lies near, but off from the 1.93–1.88 Ga segment of the APWP of Fennoscandia (Elming et al., 1993). Previous palaeomagnetic studies of the Fennoscandian shield have reported only a few formations (e.g. Keuruu dykes, ~1.88 Ga; Pesonen, 1990) with a reversed magnetization from that period (Pesonen et al., 1991; Elming et al., 1993). If the divergent position of pole C, relative to the APWP, is due to plate movement, the Fennoscandian APWP (Mertanen et al., 1989; Pesonen et al., 1989; Elming et al., 1993) should be revised to include these R poles. Based on the contact evidence, showing that the ultramafic intrusion is younger than the surrounding gabbro-diorite, and on the interpretation that C is the primary magnetization in the ultramafic intrusion, the age of magnetization C must be between 1.93 and 1.88 Ga.

The normal polarity pole A of the ultramafic intrusion (Fig. 14) has the same position as the most typical magnetization of the rocks that were formed during, or were metamorphically affected by the Svecokarelian orogeny ~1.9–1.8 Ga ago (e.g. Pesonen et al., 1989; Elming et al., 1993). Rb–Sr whole-rock (Meriläinen, 1976) and U–Pb titanite (Huhma and Meriläinen, 1991) isotopic age determinations indicate a thermal overprint and resetting of the Archaean crust also in the Archaean Lapland terrain ~1.9 Ga ago. According to Gaál et al. (1989) and Kesola (1993), the temperature of metamorphism in the Tsuomasvarri area was about 450°C, which could have partly reset the older magnetizations.

Alternatively, the acquisition of magnetization A may also be related to the cooling of the

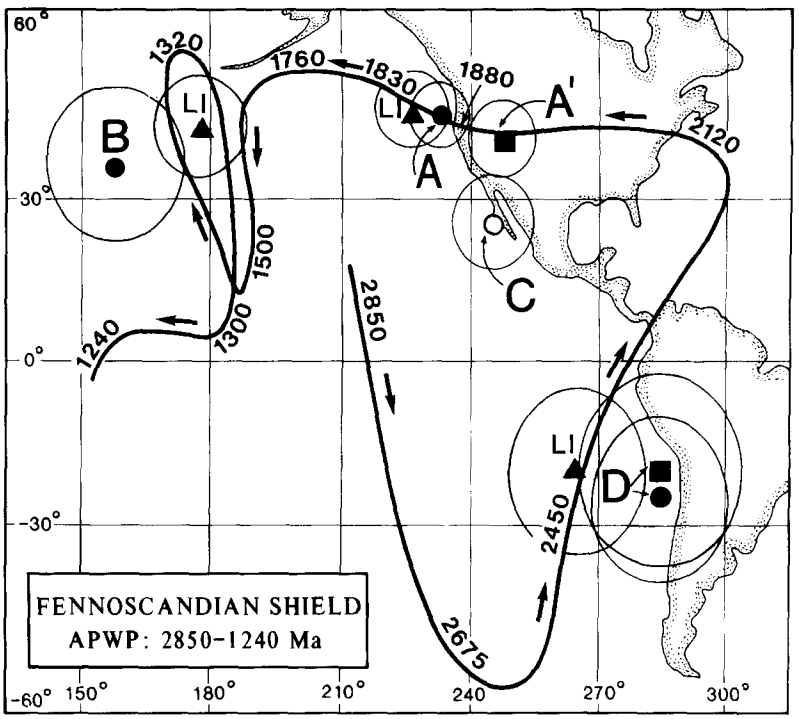


Fig. 14. APW path for the Fennoscandian shield from ~2850 to 1240 Ma (Elming et al., 1993). Grand mean poles of each magnetization are shown: dots = poles D, C, A and B from the Tsuomasvarri ultramafic intrusion; squares = poles A′ and D from the Tsuomasvarri gabbro-diorite intrusion. For comparison, grand mean poles from the Koillismaa and Koitelainen layered intrusions (Mertanen et al., 1989) are shown: *LI* ▲ = poles D, A and B. Closed (open) symbols denote normal (reversed) polarity. 95% confidence cones (A_{95}) drawn around the poles.

ultramafic magma. The absence of A in the central parts of the ultramafic intrusion, crystallized in the first stages of magma ascension as well as its absence in the gabbro-diorite intrusion, may indicate that A represents a magnetization acquired in the latest stages of the cooling ultramafic magma. On the Fennoscandian APWP, the age of A in Tsuomasvarri is about 1.88 Ga.

Pole B has about the same position on the APWP (Fig. 14) as the previously (Mertanen et al., 1989) described "B" magnetization from other Palaeoproterozoic layered intrusions, where it was interpreted as representing the last phase, about 1.75 Ga old, of the long-lasting Svecokarelian orogeny. Both B poles are, however, slightly off the latest interpretation of the APWP of the Fennoscandian shield (Elming et al., 1993), which should probably be readjusted to include these B poles. The B pole of the Tsuomasvarri intrusion is consistent with an age of, or slightly less than 1.75 Ga. The post-orogenic activity of the Svecokarelian orogeny associated with low-temperature metamorphism was probably responsible for partial resetting of the older remanences in Tsuomasvarri, giving rise to B. The same B component was recently found in the 2.3 Ga old Akanvaara gabbro farther south in the Fennoscandian shield (Mertanen et al., 1992). Elming (1990a, b) has isolated a similar B magnetization from Proterozoic intrusions in northern Sweden; he infers that it is probably associated with the Late Svecofennian generation of granitoids (~1.76 Ga) or with the slightly younger Subjotnian (1.6 Ga) activity, as sug-

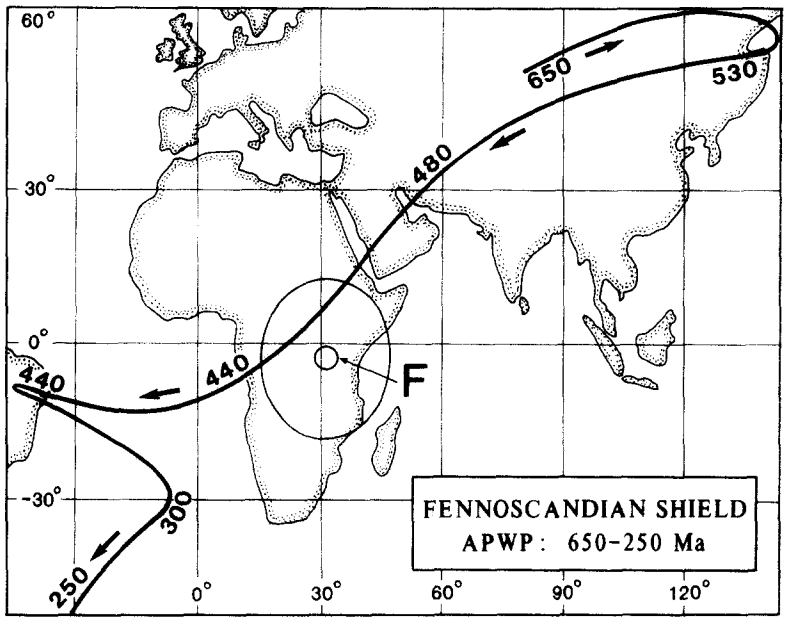


Fig. 15. APW path of the Fennoscandian shield from latest Precambrian to Permian time (650–250 Ma) (Elming et al., 1992). The polarity of the F magnetization from the Tsuomasvarri ultramafic intrusion is not known but is presented as reversed.

gested by Rb–Sr age data on rocks in northern Sweden.

Mitrofanov et al. (1991) provide Rb–Sr isotopic age determinations, also supporting metamorphism at 1.75±0.5 Ga ago in the Pechenga and Imandra–Varzuga areas (Fig. 1). From northern Finland, isotopic age determinations imply post-orogenic activity at ~1.72–1.79 Ga ago. For example, the Vainospää granite intrusion, located about 50 km southeast of Tsuomasvarri (Haapala et al., 1987) has the U–Pb isotopic age of 1.79 Ga (Huhma, 1986). Biotites from the Opukasjärvi greenstone belt adjacent to the Tsuomasvarri gabbro-diorite intrusion give K–Ar metamorphic ages of 1.70–1.81 Ga (Zwicky, 1991), supporting the interpretation that an event ~1.7–1.8 Ga ago formed B.

A similar magnetization direction corresponding to D in the Tsuomasvarri intrusion has recently been discovered in many Palaeoproterozoic rocks of the Fennoscandian shield (Fig. 14). In Finland (Mertanen et al., 1987, 1989; Mertanen and Pesonen, 1990) and in Russian Karelia (Krasnova and Gooskova, 1990), component D has been isolated in Palaeoproterozoic layered intrusions and associated dykes. The U–Pb age (zircon) of the layered intrusions in the Koillismaa complex, eastern Finland, is 2436±5 Ma (Alapieti, 1982). In all these formations, D has been interpreted as representing primary magnetization acquired about 2.4 Ga ago. Recently D has also been isolated in the 2.3 Ga old Akanvaara gabbro (Mertanen et al., 1992) lying between the Tsuomasvarri and Koillismaa intrusions (Fig. 1). In northern Sweden, the D component has been isolated in greenstones and metadiabases with a maximum age of ~2.2 Ga (Elming, 1990b). Regionally the Tsuomasvarri intrusion might be part of the same magmatic event as the layered intrusion of Imandra (isotopic age 2398 Ma, Mitrofanov et al., 1991), Mt. Generalskaya, Moncha and Panski-Fedorova (age 2490–2453 Ma, Mitrofanov et al., 1991), all of which occur in the rift-related Pechenga–Varzuga belt (Fig. 1).

However, if magnetization D is considered ~2.3–2.4 Ga old, its occurrence in the Tsuomasvarri ultramafic intrusion contradicts geological evidence according to which the ultramafic body is younger than the surrounding gabbro-diorite

intrusion dated to 1.93 Ga. We are thus left with a dilemma for the age of component D: this study favors an age of ~1.9 Ga whereas previous studies in Finland and Russia suggest an age of ~2.4 Ga. We note here that pole D does not fit the Palaeozoic Fennoscandian APWP (Elming et al., 1993) either.

The low T_{LUB} and the coercivity spectra of component F suggest that it is a secondary remanence. Its pole position is very close (but of opposite polarity) to the pole from the Laanila diabase dyke about 100 km southeast of the Tsuomasvarri formation, suggesting an age of ~1.0 Ga (Pesonen et al., 1991). However, no isotopic age data support the 1.0 Ga event in this area. Furthermore, its location on the Palaeozoic APWP (Elming et al., 1993; Fig. 15) suggests an age between 480 and 440 Ma. This age is possible for pole F, although there is no direct evidence of geological events of that age in this part of the Fennoscandian shield. However, the Caledonian front lies only about 25 km northwest of the Tsuomasvarri intrusions. According to Romer (1989), Caledonian thrusting may influence the shield up to distances of 100 km from the Caledonian front, resulting in low-temperature (up to 300°C) chemical and mechanical reactivation of the basement. That magnetization F was acquired during the Caledonian orogeny is further supported by the observation that the lower intercept point of the U–Pb concordia diagram yields an age of 399±1 Ga (M. Vaasjoki, in Forsberg-Heikkilä, 1989). Gebauer and Grunenfelder (1976) have shown that the zircon suites can be open systems at temperatures of at most 350–400°C. Such a weak thermal pulse is probably not recorded in the mineral assemblage of the host rocks, but can be seen in the isotopic and palaeomagnetic data.

8.2. Gabbro-diorite intrusion

In the gabbro-diorite, the pole position of the characteristic remanence A′ is clearly distinct from the A in the ultramafic body, and it gives a magnetization age of ~1.93 Ga (Fig. 14). The U–Pb zircon age (1.93 Ga) for the gabbro-diorite intrusion was obtained from samples collected at site TL; the site mean remanence direction corresponds well with the mean direction of the entire intrusion. Thus, the age of pole A′ can be regarded as ~1.93 Ga.

The fact that we have not found components A, B or F in the gabbro-diorite may be due to the better resistance of these rocks to later geological events since A′ was acquired. The composition of the ultramafic intrusion, on the other hand, allows more variation in the magnetic mineralogy (as shown by microscopic studies and hysteresis properties), and this is reflected in the complexity of the magnetizations in the rocks.

Pole D of the gabbro-diorite does not differ significantly from the equivalent pole D of the ultramafic body. However, component D of both intrusions is poorly determined due to the small number of samples in which D was isolated and due to considerable between-site scatter. Here, too, we face the same problem encountered with the ultramafic intrusion: the inferred APWP age for D, ~2.4 Ga (Fig. 14), does not fit the geologic and isotopic age evidence.

An alternative model for the evolution of the area (Mertanen and Pesonen, 1992) suggested that the age of the gabbro-diorite, 1.93 Ga, represents the age of regional metamorphism, not the age of emplacement. Subsequently, D could be the primary magnetization, dating ~2.4 Ga, while the others are secondary. In that case, however, the temperature required to make the U–Pb system open must have been at least 800°C (M. Vaasjoki, pers. commun., 1991). Older remanent magnetization would not be able to survive in such high temperatures. Hence, this model is untenable.

The occurrence of D in Tsuomasvarri casts now doubt on previous palaeomagnetic results from 2.4 Ga old layered intrusions and associated dykes of Finland and Russia (Fig. 1) where a similar magnetization has been interpreted to represent a primary, 2.4 Ga old, magnetization (Mertanen et al., 1989; Krasnova and Gooskova, 1990; Elming et al., 1993). The reliability grade (using the modified grading scheme of Briden and Duff, 1981, in Pesonen et al., 1989) in the Burakovskaja layered intrusion and in the Shalsky and Kalja River dolerite dykes in Russia

(Krasnova and Gooskova, 1990; Elming et al., 1993) is, however, poor (grades D and C, respectively). Only in the Burakovskaja gabbro-dolerite dykes and in the layered intrusions of Koillismaa the reliability grades are good, varying from A (Krasnova and Gooskova, 1990; Elming et al., 1993) to B (Mertanen et al., 1989), respectively. In these formations the D magnetization is most likely primary, as based on tilt and contact evidence and on thermal decay curves. D has not previously been found in rocks younger than 2.2 Ga (Pesonen et al., 1991). Naturally, when comparing coinciding poles from different areas, we should remember that in some areas the pole may represent a primary remanence, but in others a secondary one. Thus, in the Palaeoproterozoic layered intrusions and associated dykes, it is possible that the D is primary, whereas in Tsuomasvarri the similarly appearing "D" is younger.

9. Conclusions

The remanence components in the Tsuomasvarri intrusions suggest that the northern Fennoscandian shield has had a complex geological history. The following conclusions about the evolution are drawn in the light of palaeomagnetic results.

(1) The gabbro-diorite intrusion, emplaced ~1.93 Ga ago, acquired a primary thermoremanence A′ upon cooling of the magma, and the magnetization has been preserved from later overprints.

(2) Later, ~1.9 Ga ago, the ultramafic intrusion was emplaced and during cooling acquired the reversed polarity magnetization C. Considerable plate movement between these two magmatic events has taken place.

(3) The normal polarity component A is interpreted to be a thermochemical remanence, acquired ~1.88 Ga ago during the peak of the Svecokarelian orogeny or in the last stages of the long-lasting cooling of the ultramafic magma.

(4) Component B represents post-orogenic activity of the Svecokarelian orogeny, associated with low-temperature metamorphism which partly reset the older remanences in Tsuomasvarri ~1.75 Ga ago.

(5) The age of component D is unclear. On the APWP the age is 2.4 Ga. From isotopic age evidence, however, this age is aberrant. If the 2.4 Ga age of D is realistic, its occurrence in the Tsuomasvarri ultramafic intrusion may indicate that the ultramafic intrusion represents the same magmatism as the layered intrusions in Kola Peninsula.

(6) Component F is the youngest magnetization and was probably acquired as a consequence of Caledonian thrusting ~500–400 Ma ago. This overprint, acquired at low temperatures (below 350°C), is probably not distinguishable in the mineralogy of the Tsuomasvarri rocks. Thus, palaeomagnetism may be a useful method for detecting geological events that cannot be observed by other techniques.

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Appendix 1
Site mean magnetization data, Tsuomasvarri ultramafic layered intrusion (69.9°N, 28.3°E)

Site	*B/N/n*	*D* (°)	*I* (°)	α_{95} (°)	*k*
Component A					
TB	1/4	347.7	43.5	–	–
TE	1/1	335.7	40.4	–	–
TF	2/6	342.3	36.7	–	–
TG	2/2	345.3	42.8	–	–
TJ	2/3	335.2	46.9	–	–
TK	1/1	340.1	50.2	–	–
Mean	6/9/15	341.1	43.5	5.0	182.8
Component B					
TB	3/3	34.8	19.8	26.8	22.2
TF	1/2	56.8	47.5	–	–
TG	1/1	44.2	48.8	–	–
TI	1/1	45.0	39.7	–	–
TJ	1/1	37.0	33.8	–	–
Mean	5/7/9	42.7	38.2	13.0	35.9
Component C					
TA	2/7	129.5	−11.0	–	–
TB	2/7	143.4	−5.1	–	–
TD	4/18	154.4	−35.2	12.1	58.7
TE	5/28	146.6	−15.4	5.9	171.0
TF	5/28	154.5	−17.5	7.6	101.3
TG	2/7	161.0	−8.8	–	–
TH	3/9	148.3	−13.4	29.3	18.8
TI	5/22	152.9	−17.2	17.2	20.7
TJ	1/1	129.4	−19.7	–	–
TK	3/8	150.0	−33.2	20.6	36.7
Mean	10/32/135	146.9	−17.9	8.4	33.9
Component D					
TA	4/12	95.7	52.9	23.8	15.8
TE	2/6	96.2	21.5	–	–
TF	2/7	104.6	49.2	–	–
TG	1/2	118.1	51.8	–	–
TH	1/3	71.1	41.7	–	–
TI	1/2	77.5	57.6	–	–
Mean	6/11/32	93.7	47.0	14.6	21.9
Component F					
TA	1/4	164.0	42.5	–	–
TB	1/6	176.4	23.5	–	–
TD	1/5	205.1	33.1	–	–
TE	1/1	172.0	33.8	–	–
TI	1/4	164.8	22.5	–	–
Mean	5/5/20	176.3	31.9	15.7	24.6

Site locations are shown in Fig. 3. *B/N/n*=number of sites/samples/specimens in which the magnetization has been isolated. Site mean magnetization directions are calculated from the mean of samples. Mean magnetization directions of different components for the whole intrusion are calculated from the means of sites. *D*=declination; *I*=inclination; α_{95}=the radius of a cone of 95% confidence about the mean; *k*=the Fisher (1953) precision parameter.

Appendix 2
Site mean magnetization data, Tsuomasvarri gabbro-diorite intrusion (69.9°N, 28.3°E)

Site	N/n	D (°)	I (°)	α_{95} (°)	k
Component A′					
TC	2/7	329.5	38.9	–	–
TL	4/13	326.8	36.7	13.8	45.5
TM	5/16	325.3	42.3	4.9	47.7
TN	5/18	320.0	32.6	10.9	51.5
TO	2/5	334.2	41.0	–	–
TP	1/2	344.4	47.6	–	–
TQ	3/8	334.2	39.1	18.0	48.1
TR	6/13	311.5	50.1	6.7	100.9
TS[a]	2/6	288.7	38.2	–	–
Mean	8/28/82	328.2	41.4	6.3	79.0
Component D					
TM	1/1	105.1	30.6	–	–
TN	3/8	88.8	50.6	63.6	4.8
TO	1/2	89.7	38.7	–	–
Mean	3/5/11	95.3	40.2	19.0	43.2

[a]Not used in mean calculations.
For explanations, see Appendix 1.

References

Alapieti, T., 1982. The Koillismaa layered igneous complex, Finland—its structure, mineralogy and geochemistry, with emphasis on the distribution of chromium. Geol. Surv. Finl., Bull., 319: 1–116.

Alapieti, T. and Lahtinen, J., 1989. Early Proterozoic layered intrusions in the northeastern part of the Fennoscandian Shield. In: T. Alapieti (Editor), 5th International Platinum Symposium. Guide to the Post-Symposium Field Trip, August 4–11, 1989. Geol. Surv. Finl., Guide 29, pp. 3–41.

Barbey, P. and Martin, H., 1987. The role of komatiites in plate tectonics. Evidence from the Archaean crust and early Proterozoic crust in the eastern Baltic Shield. Precambrian Res., 35: 1–14.

Barbey, P., Convert, J., Moreau, B., Capdevila, R. and Hameurt, J., 1984. Petrogenesis and evolution of an Early Proterozoic collisional orogen: the granulite belt of Lapland and the Belomorides (Fennoscandia). Bull. Geol. Soc. Finl., 56: 161–188.

Bernard-Griffiths, J., Peucat, J.J., Postaire, B., Vidal, Ph., Convert, J. and Moreau, B., 1984. Isotopic data (U–Pb, Rb– Sr, Pb–Pb and Sm–Nd) on mafic granulites from Finnish Lapland. Precambrian Res., 23: 325–348.

Berthelsen, A. and Marker, M., 1986. Tectonics of the Kola collision suture and adjacent Archean and Early Proterozoic terrains in the northeastern region of the Baltic Shield. In: D.A. Galson and St. Mueller (Editors), The European Geotraverse, Part 1. Tectonophysics, 126: 31–55.

Bibikova, E.V., Tugarinov, A.I., Gracheva, T.V. and Konstantinova, M.V., 1973. The age of granulites of the Kola Peninsula. Geochem. Int., 10(3): 508–518.

Bibikova, E., Melnikov, V., Minev, S. and Lokhov, K., 1991. Lapland granulites: isotope geochemistry and geochronology. In: P. Tuisku and K. Laajoki (Editors), Metamorphism, Deformation and Structure of the Crust. Res Terrae, Publications of the Department of Geology, University of Oulu, Joint Meeting of IGCP Projects 275: Deep Geology of the Baltic/Fennoscandian Shield, 304: Lower Crustal Processes. Oulu, Finland, August 1991, Abstr., p. 4.

Briden, J.C. and Duff, B.A., 1981. Pre-Carboniferous palaeomagnetism of Europe north of the Alpine orogenic belt. In: M.W. McElhinny and D.A. Valencio (Editors), Paleoreconstruction of Continents. Am. Geophys. Union, Geodyn. Ser., 2: 137–150.

Bugge, J.A.W., 1980. The Sydvaranger type of quartz-banded iron ore, with a synopsis of Precambrian geology and ore deposits of Finnmark. Bull. Geol. Soc. Finl., 307: 15–24.

Collinson, D.W., 1983. Methods in Rock Magnetism and Palaeomagnetism. Chapman and Hall, London, 503 pp.

Day, R., 1977. TRM and its variation with grain size. J. Geomag. Geoelectr., 29: 233–265.

Day, R., Fuller, M. and Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites, grain-size and compositional dependence. Phys. Earth Planet. Inter., 13: 260–267.

Dunlop, D.J., 1983. Determination of domain structure in igneous rocks by alternating field and other methods. Earth Planet. Sci. Lett., 63: 353–367.

Elming, S.-Å., 1990a. Paleomagnetic multicomponent anal-

yses of Archaean–Early Proterozoic rocks of northern Sweden. Unpubl. manuscript.

Elming, S.-Å., 1990b. Palaeomagnetism of Precambrian basic intrusions and volcanic rocks in northern Sweden and its correlations to radiometric data. Unpublished manuscript.

Elming, S.-Å., Torsvik, T.H., Mertanen, S., Krasnova, A.F., Bylund, G., Khramov, A.N., Leino, M.A.H., Pesonen, J.L., Smethurst, M. and Terho, M., 1993. Catalogue of palaeomagnetic directions and poles from Fennoscandia: Archaean to Tertiary, 2nd issue (in prep.).

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. London, A, 217: 295–305.

Forsberg-Heikkilä, C., 1989. Tsuomasvarrin kerrosrakenteinen ultramafinen intruusio Utsjoella. Unpublished M.Sc. thesis, Department of Geology, University of Helsinki, 88 pp.

Freeman, R., von Knorring, M., Korhonen, H., Lund, C. and Mueller, St. (Editors), 1989. The European Geotraverse, 5. The Polar Profile. Tectonophysics, 162(1–2), 171 pp.

Gaál, G. and Gorbatschev, R., 1987. An outline of the Precambrian evolution of the Baltic Shield. Precambrian Res., 35: 15–52.

Gaál, G., Berthelsen, A., Gorbatschev, R., Kesola, R., Lehtonen, M.I., Marker, M. and Raase, P., 1989. Structure and composition of the Precambrian crust along the POLAR Profile in the northern Baltic Shield. Tectonophysics, 162: 1–25.

Gebauer, D. and Grunenfelder, M., 1976. U–Pb zircon and Rb–Sr whole-rock dating of low-grade metasediments example: Montagne Noire (southern France). Contrib. Mineral. Petrol., 59(1): 13–32.

Haapala, I., Front, K., Rantala, E. and Vaarma, M., 1987. Petrology of Nattanen-type granite complexes, northern Finland. Precambrian Res., 35: 225–240.

Hanski, E. and Smolkin, V., 1989. Pechenga ferropicrites and other Early Proterozoic picrites in the eastern part of the Baltic Shield. Precambrian Res., 45: 63–82.

Hanski, E. and Kesola, R., 1990. Comparative study of the Opukasjärvi and Surnujärvi greenstone belts in northernmost Finnish Lapland and the Pechenga complex, Kola Peninsula (abstr.). In: Y. Kähkönen (Editor), Proterozoic Geochemistry, Helsinki '90. Dept. of Geology, University of Helsinki, pp. 16–17.

Hanski, E., Huhma, H., Smolkin, V. and Vaasjoki, M., 1990. The age of the ferropicritic volcanics and comagmatic Ni-bearing intrusions at Pechenga, Kola Peninsula, U.S.S.R. Bull. Geol. Soc. Finl., 62: 123–133.

Huhma, H., 1986. Sm–Nd, U–Pb and Pb–Pb isotopic evidence for the origin of the Early Proterozoic Svecokarelian crust in Finland. Geol. Surv. Finl., Bull., 337, 48 pp.

Huhma, H. and Meriläinen, K., 1991. Provenance of paragneisses from the Lapland granulite belt. In: P. Tuisku and K. Laajoki (Editors), Metamorphism, Deformation and Structure of the Crust. Res Terrae, Publications of the Department of Geology, University of Oulu, Joint Meeting of IGCP Projects 275: Deep Geology of the Baltic/Fennoscandian Shield, 304: Lower Crustal Processes. Oulu, Finland, August 1991. Abstr., p. 4.

Hörmann, P.K., Raith, M., Raase, P., Ackermann, D. and Seifert, F., 1980. The granulite complex of Finnish Lapland: petrology and metamorphic conditions in the Ivalojoki–Inarinjärvi area. Geol. Surv. Finl., Bull., 308: 1–95.

Kesola, R., 1991. Taka-Lapin metavulkaniitit ja niiden geologinen ympäristö. Summary: Metavolcanic and associated rocks in the northernmost Lapland area, Finland. Geol. Surv. Finl., Rep. Invest., 107, 58 pp. (in Finnish, with English abstract and summary).

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc., 62: 699–718.

Krasnova, A.F. and Gooskova, E.G., 1990. On geodynamic evolution of Vodlozero block of Karelia in the light of palaeomagnetic data. Izv. Akad. Nauk SSSR, Fizika Zemli, 1: 103–110.

Krill, A.G., 1985. Svecokarelian thrusting with thermal inversion in the Karasjok–Levajok area of northern Baltic Shield. Nor. Geol. Unders., 403: 89–101.

Leino, M.A.H., 1979. Remanentin magnetoitumisen mittaus-laitteisto ja muinaisen magneettisen voimakkuuden määrityksistä. Unpublished M.Sc. thesis, Department of Physics, University of Helsinki, 77 pp.

Marker, M., 1985. Early Proterozoic (c. 2000–1900 Ma) crustal structure of the northeastern Baltic Shield: tectonic division and tectogenesis. Nor. Geol. Unders., 403: 55–74.

Marker, M., 1990. Tectonic interpretation and new crustal modelling along the polar profile, northern Baltic Shield. In: R. Freeman and St. Mueller (Editors), Proceedings of the Sixth Workshop on the European Geotraverse (EGT) Project: Data Compilations and Synoptic Interpretation. European Science Foundation, Einsiedeln, March 1990, pp. 9–22.

Meriläinen, K., 1976. The granulite complex and adjacent rocks in Lapland, northern Finland. Geol. Surv. Finl., Bull., 281, 129 pp.

Mertanen, S. and Pesonen, L.J., 1990. Palaeomagnetism of the Early Proterozoic layered intrusions, northern Fennoscandia. In: A.J. Parker, P.C. Rickwood and K.L. Tucker (Editors), Mafic Dykes and Emplacement Mechanisms. Balkema, Rotterdam, pp. 251–262.

Mertanen, S. and Pesonen, L.J. 1992. Paleomagnetism of the Proterozoic Tsuomasvarri Layered Ultramafic Intrusion, Northeastern Fennoscandian Shield. Rep. Q29.1/92/2, Geophys. Dep., Geol. Surv. Finl., 51 pp.

Mertanen, S., Pesonen, L.J., Huhma, H., Lähde, S., Lammi, A. and Leino, M.A.H., 1987. Palaeomagnetic and Rock Magnetic Studies of the Early Proterozoic Layered Intrusions, Northern Finland. Rep. Q29.1/87/4, Geophys. Dep., Geol. Surv. Finl., 132 pp.

Mertanen, S., Pesonen, L.J., Huhma, H. and Leino, M.A.H., 1989. Palaeomagnetism of the Early Proterozoic Layered Intrusions, Northern Finland. Geol. Surv. Finl., Bull., 347, 40 pp.

Mertanen, S., Pesonen, L.J. and Leino, M.A.H., 1992. Palaeomagnetic studies of mafic igneous intrusions in Northern Finland (in prep.).

Mitrofanov, F., Balashov, Y. and Smolkin, V., 1991. Geochronology of magmatism and metamorphism in the Pechenga structure and layered basic intrusions of the Kola Peninsula. In: P. Tuisku and K. Laajoki (Editors), Metamorphism, Deformation and Structure of the Crust. Res Terrae, Publications of the Department of Geology, University of Oulu, Joint Meeting of IGCP Projects 275: Deep Geology of the Baltic/Fennoscandian Shield, 304: Lower Crustal Processes. Oulu, Finland, August 1991, Abstr., p. 53.

Papunen, H. (Editor), 1977. Lapin ultramafiittien geologiset, petrologiset, geokemialliset ja mineralogiset tiedot. Lapin nikkeliprojektin dokumenttikokoelma. Arch. Geol. Surv. Finl., PSMT-R11-76-2.

Pesonen, L.J., 1990. The drift of Fennoscandia during the Proterozoic with special reference to the Bergslagen province, south-central Sweden. Geol. Fören. Stockholm Förh., 112: 190–196.

Pesonen, L.J., Leino, M.A.H. and Lammi, A., 1983. Paleomagnetism of the Baltic Shield—a state of art review. In: S.E. Hjelt (Editor), The Development of Deep Geoelectric Model of the Baltic Shield, 2. Department of Geophysics, University of Oulu, Rep. 8, pp. 355–370.

Pesonen, L.J., Torsvik, T.H., Elming, S.-Å. and Bylund, G., 1989. Crustal evolution of Fennoscandia—palaeomagnetic constraints. Tectonophysics, 162: 27–49.

Pesonen, L.J., Bylund, G., Torsvik, T.H., Elming, S.-Å. and Mertanen, S., 1991. Catalogue of palaeomagnetic directions and poles from Fennoscandia: Archaean to Tertiary. Tectonophysics, 195: 151–207.

Puranen, R., 1989. Susceptibilities, iron and magnetite content of Precambrian rocks in Finland. Geol. Surv. Finl., Rep. Invest., 100, 45 pp.

Romer, R., 1989. Lead mobilization during foreland metamorphism in orogenic belts: examples from northern Sweden. Geol. Rundsch., 79(3): 693–707.

Torsvik, T.H., 1986. IAPD—Interactive Analyses of Paleomagnetic Data (User Guide and Program Description). Institute of Geophysics, University of Bergen, 74 pp.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and S. Runcorn (Editors), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.

Zwicky, C.N., 1991. Mineralogisch-Petrographische Untersuchungen und K/Ar-Alterbestimmungen in der Opukasjärvi-Schieferzone (Nordfinnland). Unpubl. Lic. Thesis, Philosophisch-Naturwissenschaftliche Fakultät der Universität Bern, 104 pp.

Zykov, S.I., Stupnikova, N.I. and Sidorenko, S.A., 1984. Lead isotopes of rocks of the Granulite Belt of the Kola Peninsula. Moscow Univ. Geol. Bull., 39(3): 61–66. (ref. Gaál et al., 1989).